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Prediction of cardiovascular disease by using a genetic algorithm over a stacked ensemble model

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Abstract

A two-layer customized ensemble model is suggested in this work to improve the prediction accuracy of cardiovascular disease. Two ensemble classifiers are stacked as part of the model architecture. While Support Vector Machine (SVM) and Decision Tree (DT) are used as base learners in the first ensemble, and Random Forest (RF) and Logistic Regression (LR) are used in the second ensemble. The dataset used in training the model has been collected from the UCI data repository. The most pertinent attributes were chosen using LASSO (Least Absolute Shrinkage and Selection Operator), which lowers dimensionality. To further enhance the model's performance, the Genetic Algorithm (GA) has been used. The suggested model's excellent efficacy in detecting cardiovascular disease has been demonstrated by its 88.52% accuracy rate. 10 Fold Cross Validation has been added for further validation of the model and it has also been identified statistically significant over Decision Tree and Support Vector Machine classifier as per the tested results. For better understanding of prediction the proposed model has been explained using SHAP. With a strong and precise prediction framework for medical practitioners, this hybrid ensemble technique shows great promise in medical diagnosis systems.

Keywords Cardiovascular disease, Decision tree, Genetic algorithm, LASSO, Logistic regression, Random forest, SHAP, Support vector machine

1 Introduction

Machine learning is a subset of Artificial Intelligence that can analyse a huge set of data and help to make decisions without human intervention. The capability of analysing and interpreting a vast set of data makes machine learning an indispensable support in different areas where fast and accurate decision has to be made. The healthcare industry is such a field where machine learning is currently playing a very crucial role, from disease prediction to providing an end-to-end assistant to patients. The research objective of this work is to focus on the machine learning application in disease prediction.



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Cardiovascular diseases persist as a predominant cause of mortality globally, rendering early and precise diagnosis an essential objective in healthcare research. As clinical data becomes more accessible, machine learning has become a useful tool for making predictive systems that can help doctors find diseases and figure out how likely someone is to get sick. Even though algorithmic methods have gotten better, traditional single-model classifiers still have a hard time finding the right balance between accuracy, generalization, and robustness, especially when working with complex, nonlinear medical data.

Prior investigations in medical diagnostics have examined an extensive array of machine learning methodologies, encompassing Support Vector Machines (SVM), Logistic Regression, Decision Trees, and Random Forests. Logistic Regression: Easy to interpret, less likely to overfit, scalable and fast. Decision Tree: Also easy to interpret and normalisation of data is not required. Random Forest is a robust model that is not prone to overfitting and can handle missing data. The Support Vector Machine is robust and performs well on small to medium datasets.

Probabilistic outputs of Logistic Regression, structural rules of Decision Trees, ensemble averaging of Random Forests, hyperplane margins of Support Vector Machines. The final meta-model is accurate, robust to outliers and able to generalise successfully across different data sets.

This study's main research question is how to build a classification model that keeps its predictive accuracy high while lowering overfitting and making it easier to generalize to new medical data. This study specifically aims to improve the prediction of heart disease, a field where minor inaccuracies can lead to significant consequences in reality.

The objective of this research is to develop a resilient and flexible classification framework that amalgamates sophisticated ensemble learning with evolutionary optimization methods to enhance diagnostic precision. The study posits that the integration of various learning paradigms, coupled with the optimization of their performance parameters via evolutionary computation, will produce a model that is both precise and robust in medical prediction tasks.

This study is significant as it aims to connect computational advancements with clinical relevance. By making models more reliable and easier to understand, these kinds of systems can help healthcare workers make quick, confident, and data-driven decisions about diagnoses. Moreover, the employment of optimization-driven ensemble methods signifies progress in the utilization of artificial intelligence in healthcare, highlighting adaptability, accuracy, and resilience in predictive modelling.

The proposed method integrates GA-based wrapper feature selection and hierarchical weighted soft voting ensemble to obtain the most informative clinical features and leverage the complementary strengths of Support Vector Machine, Decision Tree, Random Forest, and Logistic Regression classifiers. Such architecture enhances the robustness of the model and decreases the redundancy of features compared with the conventional single classifier approach. It leverages also many learning paradigms to improve generalisation. Rather than a single train-test split, we performed a thorough validation of the system using stratified 10-fold cross-validation and paired statistical significance testing to ensure the robustness and reproducibility of the results. In addition to the traditional evaluation criteria, the suggested work includes computational complexity analysis, confusion matrix analysis, ROC-AUC evaluation and robustness assessment for a comprehensive performance assessment. The proposed framework is a statistically validated,

interpretable and computationally efficient decision-support system for the prediction of heart disease, which makes it more suitable for real world clinical applications where reliability, stability and reproducibility are as important as maximising the predictive accuracy.

In Sect. 2, some of the proposed works in this field have been analysed, and the proposed methodology has been represented in detail in Sect. 3. Experimental results have been presented in Sect. 4, followed by a conclusion in Sect. 5.

2 Literature review

Alanazi proposed a model for the prediction of chronic disease. K-Nearest Neighbour (KNN) has been used for distance computation to discover the exact match in the data set and the final disease prediction output. Convolutional Neural Network (CNN) has been employed for automatic feature extraction and disease prediction [1]. In order to bridge the gap between patients and doctors to reach their own goals, a system that can predict different diseases with the use of several machine learning algorithms, such as Naïve Bayes, KNN, DT, random forest, and SVM algorithms, has been reported in [2].

By allocating weights according to their rarity, the model proposed by Gaurav et al. [3] alters the medical dataset that it uses from Kaggle. Then, a combination of Random Forest, Long Short-Term Memory (LSTM), and SVM machine learning techniques is used to train this dataset. In parallel with this, the patient's history can be examined using the LSTM Algorithm. Next, SVM is employed to determine the potential ailment. Comparing the suggested model to cutting-edge techniques, more accuracy and dependability have been achieved. The suggested model is beneficial for advancing the automation of the healthcare sector.

The KNN algorithm, which was employed by Keniya et al. [4]. They applied this strategy, which is sensitive to noisy and missing data, by classifying the data point into the group to which the majority of the K data points belong. In order to anticipate the condition, they have taken into account a number of variables, including the patient's gender, age group, and symptoms. By utilizing the Naïve Bayes approach, Pingale et al. [5] have developed a strategy for forecasting specific diseases, including diabetes, malaria, jaundice, dengue, and tuberculosis. They haven't used a sizable dataset to forecast a sizable number of illnesses.

The Rustboost Algorithm was employed in the model proposed by Kumar et al. [6]. The purpose of RUSBoost is to alleviate the problem of class imbalance. However, the resampling technique used by the RUSBoost algorithm, random under-sampling, may result in the loss of important data. As a result, when training the data, this technique was ignored. The Naïve Bayes classifier was employed in the model proposed by Chhogyal and Nayak. They have not been using the conventional dataset for training, and their performance in predicting diseases has been poor [7].

Chen et al. [8] optimised a number of machine learning techniques to predict a chronic disease epidemic effectively. The information gathered for the training purpose was insufficient. A latent component model was applied in order to get over this. An optimized ensemble model was developed by blending a DNN with LightGBM and XGBoost machine learning models to predict diseases using laboratory test results. From a dataset comprising 5145 cases and 326,686 test results, 86 attributes were selected based on value counts, clinical importance, and missing values. The focus was on 39 diseases

based on ICD-10 codes. The ensemble model achieved an F1-score of 81% and a prediction accuracy of 92% for the five most common diseases [9]. Zengchen Yu et al. proposed a model applying Artificial Neural Network, FM-Deep Learning, CNN, and Recursive Neural Network (RNN). The integration of digital twins and the advancement of precision medicine—the two main trends in the field of medicine and disease prediction—are explained in this work [10].

A model that contrasted several models—including RF, XGB, MLP, ARIMA, ARIMAX, GLARMA, and SARIMA—was put forth by Dixon S. et al. In the end, tree-based machine learning models outperformed other models when examining data splits, with XGB models outperforming other models for all diseases. The study, according to the authors, showed how machine learning techniques may integrate a variety of parameters to more precisely forecast different diseases, independent of geography, than can be achieved using traditional statistical approaches [11].

By combining an Ordinary Differential Equation (ODE) model for infectious classes with a Gradient Boosting Machine (GBM) to predict how public health policies and mobility data influence the transmission rate in the ODE model, Wang X. et al. developed a method to predict the daily number of confirmed cases of infectious diseases [12]. Shaghghi et al. showed that 89% accuracy in predicting the course of the COVID-19 pandemic in the United States was achieved by using eVision, an epidemic prediction system that combines machine learning in the form of a RNN, LSTM and search engine statistics to make accurate predictions of the weekly number of cases of highly communicable diseases [13].

The predictions of a computational model are combined with machine learning, a projection model, and a suggested smoothed endemic channel calculation in the technique presented by González-Bandala et al. [14]. The forecasts are based on weekly data on acute respiratory infections (ARIs) gleaned from Mexican epidemiological reports and the use of search keywords in Google. When this methodology's results were compared to those of cutting-edge approaches, MAPE (maximum absolute percent error) was achieved at 21.7% and Root-Mean-Square Percentage Error (RMPSE) was decreased.

Zhang et al. developed a model whose experimental findings imply that it is possible to use Twitter data to offer COVID-19 surveillance and prediction in the US to help health departments' decision-making [15]. Bae et al. projected the impact of social media on the transmission of COVID-19 using a Monte Carlo technique and a modified Susceptible Exposed Infectious Recovered (SEIR) model to illustrate the efficacy of information shared on social media [16].

Chimmula et al. forecasted the COVID-19 pandemic utilising past transmission records. An extra variable included in the model for forecasting the illness outbreak is the COVID-19 patient recovery rate. In this work, the previous transmission data is incorporated into the prediction method using an LSTM model [17]. Chandra et al. used an encoder-decoder LSTM model, a bi-directional LSTM model, and a LSTM model to predict the spread of COVID-19 in a few Indian states. States that have shown an improvement in infection rates are contrasted with those that have the highest infection rates [18].

Hasan and Bao conducted a comparative analysis of several algorithms to determine which feature selection method was most effective in forecasting cardiovascular disease. To obtain feature subsets, they combined filter, wrapper, and embedding techniques

using a Boolean procedure. XGBoost, SVC, k-nearest neighbours, random forest, and Naive Bayes were among the models that were assessed. The best accuracy was achieved by XGBoost using the wrapper approach (73.74%), SVC (73.18%), and ANN (73.20%) [19]. Machine learning technologies for the identification of heart disease were proposed and analysed by Umarani Nagavelli et al. To predict cardiac illness, they first employed a weighted Naïve Bayes algorithm. Second, they used information theory, time domain, and frequency domain variables to identify ischemic heart disease. They used SVM and XGBoost as classifiers. Third, using an enhanced SVM based on duality optimisation, they were able to identify heart failure. Finally, they created a Heart Disease Prediction Model (HDPM)-based Clinical Decision Support System (CDSS) that uses XGBoost for prediction, SMOTE-ENN for data balance, and DBSCAN for outlier detection [20].

Temporal Association Rules (TARs) were employed in a Naïve Bayes classification model by Kalia Orphanou et al. to diagnose heart disease. To detect TARs, they employed a temporal pattern mining technique after preprocessing the data with Temporal Abstraction (TA). When the recurrence of each TAR pattern was taken into account together with pertinent medical information, these periodic TARs were utilised as features, and an accuracy of 82% was achieved [21]. Using machine learning techniques including Principle Component Analysis (PCA), univariate feature ranking, and a multiple logistic regression classifier, a model was developed to determine who is most at risk for CVD. The study found that the most important clinical factors were duration of diabetes, plaque scores, and hypercholesterolemia. With an AUC of 0.87, the machine learning method successfully identified 40/47 (85.11%) high-risk patients and 114/144 (79.17%) low-risk patients. Based on basic patient criteria, a machine learning technique is helpful in identifying MAFLD patients with extensive CVD, according to the study's findings [22].

The goal of Alotalibi et al.'s study [23] was to ascertain how well machine learning methods might predict the development of heart failure. Using a dataset from the Cleveland Clinic Foundation and a variety of machine learning approaches, such as decision trees, logistic regression, random forests, naive Bayes, and support vector machines, they created prediction models. When developing the model, they used a 10-fold cross-validation technique. The outcome demonstrated that the decision tree algorithm performed better in predicting heart disease than the SVM algorithm, with an accuracy rate of 93.19%.

A customised ensemble classifier for the categorization of heart disease was proposed by AB. Majumder et al. The accuracy, specificity, and sensitivity of each base classifier were taken into consideration while determining the weights of the ensemble classifier. The Chi-Square method was utilised for feature selection, while the Firefly algorithm was employed to enhance model performance. The suggested model's accuracy was 85.52% [24].

Majumder et al. have introduced a novel way to vote classification: concatenated hybrid ensemble voting [25]. This innovative method combines two classifiers, with each hybrid ensemble classifier using a distinct set of fundamental classifiers, including Support Vector Machine, Decision Tree, K-Nearest Neighbour, Logistic Regression, Adaboost, and Naive Bayes. Through the use of concatenated ensemble classifiers, the proposed model demonstrates promising performance results, most notably achieving an accuracy rate of 86.89%. This approach highlights how prediction models may be

improved in a variety of domains and shows how ensemble learning approaches can be used to increase classification accuracy.

According to research by Apurb Rajdhan et al., with an accuracy rate of 90.16%, the Random Forest algorithm was the best in predicting cardiac ailments. Within the UCI Cleveland Heart Disease dataset, which contained 76 features, they chose 14 features. Age, sex, maximal heart rate, and the degree of chest discomfort are some of these features. They not only tried Random Forest but also Naive Bayes, Decision trees, and Logistic Regression; the accuracy for each was 85.25%, 85.25%, and 81.97%, respectively [26].

By utilising a variety of machine learning methodologies and techniques, Albert Alexander Stonier et al. [27] presented a model that forecasts the likelihood of a heart attack. In order to improve medical diagnosis, the study looks at a number of techniques, such as Random Forest, Regression models, K-nearest neighbour imputation, Naïve Bayes algorithm, etc. The discovery that the Random Forest algorithm can predict the risk of a heart attack with an improved accuracy of 88.52% could herald a revolution in the diagnosis and treatment of cardiovascular disorders.

Subramani S et al. created a model for predicting heart illness using several machine learning approaches. A gradient boosting decision tree was used in this work to pick features. Among the several methods employed were Naïve Bayes, Decision Tree, Support Vector Machine, Logistic Regression, K Nearest Neighbour, and Random Forest. An average AUC score of 0.88 was obtained in [28].

Senthilkumar Mohan et al.'s Hybrid Random Forest with Linear Model (HRFLM) combines a random forest and a linear method [29]. Compared to other algorithms studied, HRFLM's predictions were determined to be the most accurate, exhibiting the fewest errors. HRFLM's accuracy rating was 88.7%.

Haricharan Putlagunta proposed a model of heart disease prediction using Logistic Regression, Decision Tree, Random Forest and Support Vector Machine (SVM). These models are implemented and compared. Data pre-processing steps such as handling missing values, Feature scaling and Exploratory Data Analysis (EDA) were done to improve model performance. The models were evaluated on standard metrics including accuracy, precision, recall and F1-score. Random Forest algorithm performed the best in terms of accuracy and overall performance among all the models [30].

Tanzila Kehkashan et al. proposed a robust deep learning based model for heart disease prediction using 1D CNN. The dataset was collected from Kaggle. LIME and SHAP have applied for model interpretation[31].

Arpita Chakraborty et al. worked on collected clinical dataset of 5000 patients. Synthetic Minority Oversampling Technique (SMOTE) has been applied to handle data imbalance. Logistic Regression, Decision Tree, Support Vector Machine and Random Forest were tested using 10-fold stratified cross-validation and for model interpretation SHAP has been applied [32].

3 Proposed method

The proposed model has been designed by applying a hybrid customized ensemble stacking model optimized through a genetic algorithm. The dataset has been collected from the UCI heart disease data repository. For feature selection, LASSO has been applied. Initially, two ensemble weighted voting classifiers have been developed at level

1. The first ensemble classifier has been developed utilizing Support Vector Machine and Decision Tree as base learners whereas the second ensemble classifier has been designed applying Random Forest and Logistic Regression as base learners. The weight of the classifier has been assigned based on the accuracy of each base classifier and voting mechanism used is soft voting. In level 2, a Stacking ensemble classifier has been designed applying the above-mentioned two weighted voting classifier as base learner as Logistic Regression as Meta Learner. After that, the performance of the proposed hybrid customized ensemble classifier has been optimized by applying the genetic algorithm. The flow diagram of the proposed model is shown in Fig. 1.

A stacked ensemble classifier has been designed where two base learners have been used, which are also an ensemble model. The first base learner has been designed utilizing Support Vector Machine and Random Forest through a weighted voting technique. For the second ensemble classifier, Random Forest and Logistic Regression have been applied, and this is also through the weight voting technique. In both cases, accuracy has been considered as a weight. The model has been trained with the heart disease dataset collected from the UCI data repository. Originally, the dataset consisted of fourteen features, which have been reduced to eight features by applying LASSO for feature selection. The reason behind choosing LASSO as a feature selection method is that it helps to reduce overfitting and is interpretable itself. The number of features that are preserved in LASSO is determined by a penalty factor, and selecting the penalty factor through cross-validation increases the probability that the model will perform well in future data sets. The predictions of several base learners are combined in ensemble models to create a more reliable and accurate model. The effectiveness of the ensemble model can be improved by utilizing various base learner types, including Support Vector Machines (SVM), Decision Trees, Logistic Regression, and Random Forests. SVMs perform well in high-dimensional domains and, thanks to kernel techniques and regularization capabilities, are resistant to overfitting. Decision trees are easy to understand, straightforward, and don't require feature scaling to handle non-linear relationships. For binary classification, logistic regression is the best option since it is effective, simple to understand,

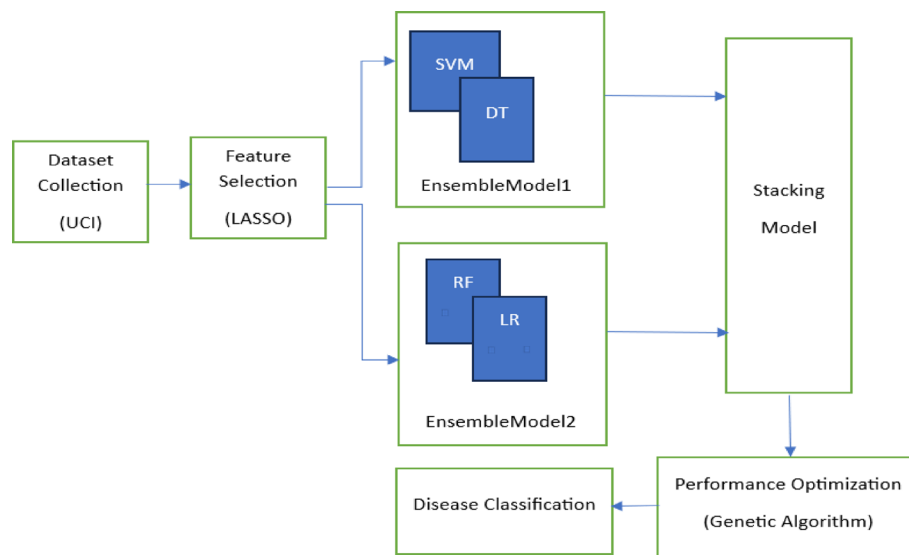


Fig. 1 Block diagram of the proposed hybrid customized ensemble stacking model optimized through a genetic algorithm

and yields probabilistic predictions. Through the aggregation of numerous trees, Random Forests provide tolerance to noise, handle missing values, and provide insights into feature relevance while reducing overfitting. By combining these various models in ensemble techniques like boosting, stacking, and bagging, their distinct advantages are tapped into, improving predictive modeling's generality, accuracy, and robustness. Predictive performance and resilience are greatly improved by stacking two base learners, each of which is an ensemble technique. This method makes use of the advantages of many ensemble techniques, such as boosting and bagging, to produce a meta-learner that is more potent. Enhanced variety, decreased overfitting, and the ability to capture intricate data correlations that separate models could overlook are the advantages of the combined model. This approach also offers flexibility in model selection and enhances generalization to previously unseen data, making it applicable to a wide range of issues. The goal of this post-stacking optimization is to minimize overfitting, increase generalization, and enhance predictive accuracy by balancing the contributions of each base learner. Because GA is evolutionary in nature, it can investigate a wide variety of possible solutions, which strengthens and adapts the final model to complex medical data. As a result, this combination produces a very accurate and dependable disease classification system that can provide an accurate medical diagnosis.

This work combines evolutionary optimization with advanced ensemble techniques, pushing the limits of classic machine learning approaches in healthcare and producing a potent tool for disease categorization. The resultant model is an effective, comprehensive solution that can be used to diagnose medical conditions in the real world. It helps medical practitioners make trustworthy and precise judgments, which in turn improves patient outcomes.

3.1 Dataset descriptions

The proposed model has been trained with the dataset collected from the UCI data repository. Age, Sex, Cp, Tresbps, Chol, Fbs, Restecg, Thalach, Exang, Oldpeak, Slope, Thal, Ca, and Target are the fourteen features that make up the collected data. The histogram of the collected dataset is shown in Fig. 2.

3.2 Feature selection applying LASSO

In machine learning models, selecting features is essential as insignificant features might have a detrimental effect on functioning. Selected for its capacity to minimize overfitting, the most important characteristics in the suggested model are determined by the Least Absolute Shrinkage and Selection Operator (LASSO) Regression. To do this, LASSO finds the best features by reducing the coefficients of less important features to zero.

Consider the dataset with D instances and f features. The dataset can be represented as in Eq. (1).

$$Dataset = \{(x_i, y_i)\}_{i=1}^D \quad (1)$$

where $x_{i1}, x_{i2}, \dots, x_{if}$, are the independent feature set and y_i is the dependent feature set.

Coefficients are $\partial = \{\partial_1, \partial_2, \dots, \partial_n\}$. These coefficients in a linear model have to be calculated.

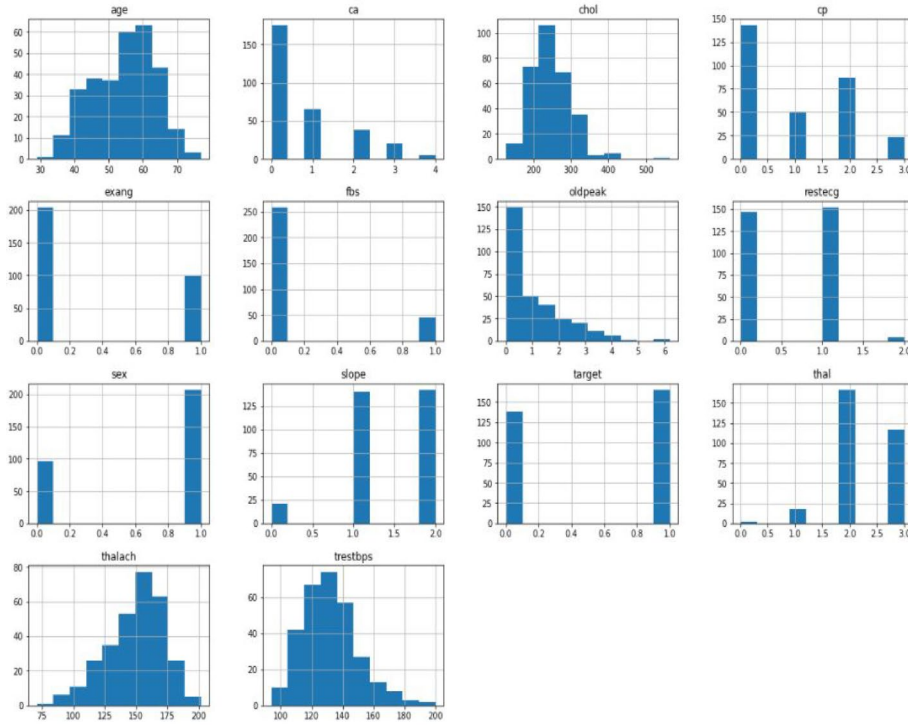


Fig. 2 Histogram of the collected heart disease dataset from the UCI data repository

$$y_i = \partial_0 + \sum_{j=1}^f \partial_j x_{ij} + \epsilon_i \quad (2)$$

where ∂_0 is the intercept and ϵ_i is the error.

LASSO objective function can be represented as in Eq. (3)

$$\min_{\partial_0, \partial} = \frac{1}{2N} \sum_{i=1}^N \left(y_i - \partial_0 - \sum_{j=1}^f \partial_j x_{ij} \right)^2 + \gamma \sum_{j=1}^f |\partial_j| \quad (3)$$

where $\frac{1}{2N} \sum_{i=1}^N (y_i - \partial_0 - \sum_{j=1}^f \partial_j x_{ij})^2$ is the residual sum of squares.

$\gamma \sum_{j=1}^f |\partial_j|$ is the L1 penalty term, and $\gamma \geq 0$ is the regularization parameter.

The L1 penalty term has the effect of shrinking some of the coefficients ∂_j exactly to zero when γ is sufficiently large. Because of this characteristic, features with non-zero coefficients are chosen to be included in the model, while those with zero coefficients are eliminated, allowing LASSO to perform feature selection.

Feature selected based on Eq. (4) below.

$$X_{selected(LASSO)} = \{x_j : \partial_j \neq 0 \text{ where } j = 1, 2, \dots, N\} \quad (4)$$

The model included seven pertinent dependent features: sex, type of chest pain (cp), thalassemia (thal), maximum heart rate attained (thalach), exercise-induced angina (exang), ST depression induced by exercise in comparison to rest (oldpeak), and the number of major vessels colored by fluoroscopy (ca). By emphasizing these characteristics, the model hopes to enhance accuracy and dependability in heart disease prediction. The histogram of the new feature set is shown in Fig. 3.

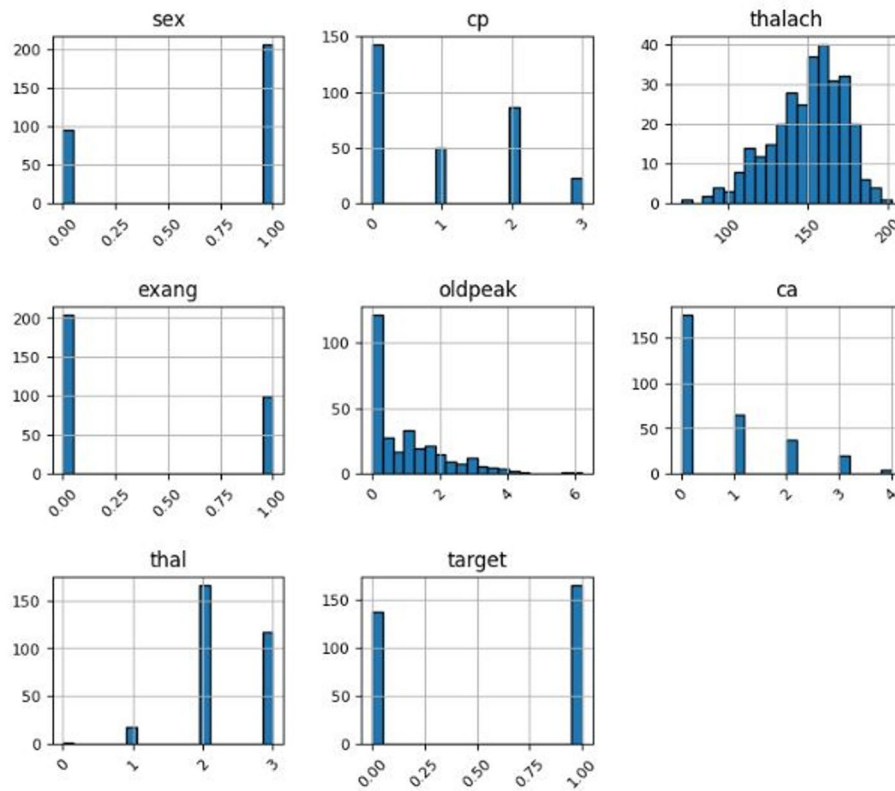


Fig. 3 Histogram of the new dataset generated by applying LASSO

3.3 Designing the ensemble Model 1

The first ensemble classifier has been designed by utilizing a Support Vector Machine and a Decision Tree as base learners through a weighted voting mechanism. The weights of the learners have been calculated based on the achieved accuracy.

Consider a training dataset (X, y) where X belongs to $X \in \mathbb{R}^{f \times n}$ and y belongs to $y \in (1, 0)$.

The goal of SVM is to locate a hyperplane $w \cdot x + b = 0$ that boosts the difference in margin between two classes.

$$\min_{w,b} \frac{1}{2} \|w\|^2 \partial \sum_{i=1}^n \max(0, 1 - y_i (w \cdot x_i + b)) \quad (5)$$

Output can be calculated based on Eq. (6).

$$f_svm(x_i) = \text{sign}(w \cdot x + b) \quad (6)$$

Using a recursive partitioning technique, DT builds a tree structure in which every node denotes a choice made in response to a feature.

Beginning at the root node, categorize a fresh sample by moving through the tree according to the feature values. Compare each node's pertinent feature. reach the threshold θ and proceed to the corresponding child node on the left or right.

The prediction function of each node can be represented as in Eq. (7).

$$f_{dt}(x, t) = \begin{cases} f_{dt}(x, t_{left}) & \text{if } x_j \leq \theta \\ f_{dt}(x, t_{right}) & \text{if } x_j > \theta \end{cases} \quad (7)$$

t is the current node t_{left} and t_{right} are the left and right child nodes, and θ is the threshold.

Leaf node can be calculated as in Eq. (8)

$$f_{dt}(x, t) = y_i \quad (8)$$

The final prediction is done by applying the $f_{dt}(x, t)$ recursively referring to Eq. (9).

$$f_{dt}(x) = f_{dt}(x, root, node) \quad (9)$$

Applying Support Vector Machine and Decision Tree a Weighted Voting classifier has been designed, for which the weights of each classifier have been calculated as in Eqs. (10) and (11).

$$W_{svm} = \frac{Accuracy_{svm}}{Accuracy_{svm} + Accuracy_{dt}} \quad (10)$$

$$W_{dt} = \frac{Accuracy_{dt}}{Accuracy_{svm} + Accuracy_{dt}} \quad (11)$$

The weighted Ensemble prediction can be represented as in Eq. (12).

$$f_{ensemble1} = W_{svm} \cdot f_{svm}(x) + W_{dt} \cdot f_{dt}(x) \quad (12)$$

3.4 Dataset splitting for training and testing

After feature selection the total dataset has been split for Training and Testing purposes. 75% of data has been used for training and 25% data has been used for testing as programming screenshot shown in the Fig. 4 below.

3.5 Designing the ensemble Model 2

In this phase, the second ensemble classifier has been designed using Random Forest and Logistic Regression, applying a weighted voting method where the weights of each classifier are calculated based on the accuracy score.

Random Forest uses majority voting to integrate forecasts from several decision trees to produce reliable and accurate predictions. Bootstrap sampling can be represented as in Eq. (13).

$$D_b = \{(x_{b1}, y_{b1}), (x_{b2}, y_{b2}), \dots, (x_{bn}, y_{bn})\} \text{ where } b = 1, 2, \dots, B \quad (13)$$

Final classification can be represented as in Eq. (14).

$$\hat{y} = \max \sum_{b=1}^B R(Predict_b(x) = y) \quad (14)$$

$R(Predict_b(x) = y)$ can be represented through Eq. (15).

```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size = 0.25, random_state = 0)
```

Fig. 4 Dataset split-screenshot

$$P(f_{rf}(x) = y) = \begin{cases} 1 & \text{if } Predict_b(x) = y \\ 0 & \text{if } Predict_b(x) \neq y \end{cases} \quad (15)$$

In the final logistic regression prediction, the class label is determined by first utilizing a decision rule to calculate the likelihood that the input belongs to class 1 using the sigmoid function. Sigmoid function can be represented as in Eq. (16).

$$p = \partial(z) = \frac{1}{1 + e^{-z}} \quad (16)$$

z can be represented as in Eq. (17).

$$z = b_0 + \sum_{i=1}^p b_i \cdot x_i \quad (17)$$

$$\hat{y} = \begin{cases} 1 & \text{if } p \geq 0.5 \\ 0 & \text{if } p < 0.5 \end{cases} \quad (18)$$

p can be represented as follows in Eq. (19).

$$p = \frac{1}{1 + e^{b_0 + \sum_{i=1}^p b_i x_i}} \quad (19)$$

Applying Random Forest and Logistic regression, a weighted voting classifier has been designed for which the weights of each classifier have been calculated as in Eqs. (20) and (21).

$$W_{rf} = \frac{Accuracy_{rf}}{Accuracy_{rf} + Accuracy_{lr}} \quad (20)$$

$$W_{lr} = \frac{Accuracy_{lr}}{Accuracy_{rf} + Accuracy_{lr}} \quad (21)$$

The weighted Ensemble prediction can be represented as in Eq. (22).

$$f_{ensemble2} = W_{rf} \cdot f_{rf}(x) + W_{lr} \cdot f_{lr}(x) \quad (22)$$

3.6 The design of stacking ensemble classifier

With the above-mentioned two hybrid ensemble classifiers, a stacking model has been designed where Logistic Regression has been used as a meta-learner. The formulated model can be represented by Eq. (23).

$$f_{stack}(y|x) = MET_{LR}(\{f_{ensemble1}(y|x)\}, \{f_{ensemble2}(y|x)\}) \quad (23)$$

3.7 Apply genetic algorithm

In this phase, the genetic algorithm with Population size = 80, mutation rate = 10%, crossover rate = 100% number of generations = 10 has been applied for performance enhancement. The processing steps of the genetic algorithm are listed below.

Let consider X be the total feature set as

$$X = \{x_1, x_2, \dots, x_n\} \quad (24)$$

Each chromosome c represents a binary vector that encodes a subset of features.

$$c = \{c_1, c_2, \dots, c_n\} \text{ where } c_i \in \{0 : \text{feature not selected}, 1 : \text{feature selected}\} \quad (25)$$

a. Initialization:

Population: A population of chromosomes, each of which represents a subset of traits, is randomly initialized.

$$\vec{p} = \{c_1, c_2, \dots, c_k\} \quad (26)$$

Sample code snippet has been added in the Fig. 5 below.

b. Fitness function: With each chromosome representing a feature subset, train and assess the stacking ensemble model.

The fitness function has been represented in Eq. (27).

$$Fit(c) = Accuracy(f_{stack}(y|x_c)) \quad (27)$$

Sample code snippet for Fitness function has been shown in Fig. 6 below.

c. Selection: Based on fitness scores, determine which K chromosomes are the best.

The selection probability for the i th chromosome is calculated as follows:

$$p_i = \frac{Fit(c_i)}{\sum_{j=1}^n Fit(c_j)} \quad (28)$$

d. Crossover: Combine segments of chosen chromosomes to create new ones. In this proposed model, single-point crossover has been used.

Sample code snippet for crossover has been shown in the Fig. 7 below.

e. Mutation: Flip bits in chromosomes at random with a specific mutation probability.

For every gene g mutation occurs in a chromosome with a probability denoted as P .

$$\hat{g}_i = \begin{cases} 1 - g_i & \text{where } rand(.) > P \\ g_i & \text{otherwise} \end{cases} \quad (29)$$

Sample code snippet for mutation has been shown in the Fig. 8 below.

f. Iteration: For a predetermined number of generations or until convergence, repeat the stages of selection, crossover, and mutation.

```
# Initialize population with random feature subsets
def initialization_of_population(size, n_feat):
    return [np.random.choice([0, 1], size=n_feat, replace=True) for _ in range(size)]
```

Fig. 5 Sample code snippet of initialization of population

```

# Evaluate fitness of population
def fitness_score(population, X_train, X_test, Y_train, Y_test):
    scores = []
    all_predictions = []
    all_probabilities = []
    for chromosome in population:
        selected_features = np.where(chromosome == 1)[0]
        if len(selected_features) == 0: # Avoid empty feature subset
            scores.append(0)
            all_predictions.append(np.zeros(len(Y_test)))
            all_probabilities.append(np.zeros(len(Y_test)))
            continue

        logmodel = VotingClassifier(estimators=[
            ('lr', LogisticRegression(max_iter=1000)),
            ('dt', DecisionTreeClassifier()),
            ('svc', SVC(probability=True))
        ], voting='soft')

        logmodel.fit(X_train[:, selected_features], Y_train)
        predictions = logmodel.predict(X_test[:, selected_features])
        probabilities = logmodel.predict_proba(X_test[:, selected_features])[:, 1]
        scores.append(accuracy_score(Y_test, predictions))
        all_predictions.append(predictions)
        all_probabilities.append(probabilities)

    return scores, all_predictions, all_probabilities

```

Fig. 6 Sample code snippet of fitness score

```

def crossover(parents, size):
    next_gen = []
    for _ in range(size):
        parent1, parent2 = np.random.choice(range(len(parents)), 2, replace=False)
        cross_point = np.random.randint(1, len(parents[0])-1)
        child = np.concatenate((parents[parent1][:cross_point], parents[parent2][cross_point:]))
        next_gen.append(child)
    return next_gen

```

Fig. 7 Sample code snippet of crossover

```

def mutation(population, mutation_rate):
    for chromosome in population:
        if np.random.rand() < mutation_rate:
            mutation_point = np.random.randint(len(chromosome))
            chromosome[mutation_point] = 1 - chromosome[mutation_point]
    return population

```

Fig. 8 Sample code snippet of mutation

g. Output: Best Feature subset.

After finishing, the GA gives the best chromosome c^* , which is the best feature subset:

$$c^* = \text{Max}(Fit(c_i)) \quad (30)$$

4 Result analysis and discussion

A hybrid customized ensemble stacking model with LASSO feature selection and the dataset from the UCI heart disease repository has been proposed. The model has been optimized through the use of a genetic algorithm. First, two weighted voting ensembles with weights depending on accuracy have been built. The first ensemble utilized SVM and Decision Tree, while the second used Random Forest and Logistic Regression. Next, using these two ensembles as base learners and Logistic Regression as the meta-learner, a stacking ensemble classifier has been constructed. In this section, the performance of the proposed model has been analysed in detail with different performance metrics. Accuracy, Sensitivity, Specificity Precision, F1-score, AUC score, Cohen Kappa and Matthews Correlation Coefficient are the performance measures which has been considered. 10 Fold cross validation has been applied over the experiments for getting detailed performance insight of the proposed model. The robustness of the model has been tested by identifying p values statistical measure. The performance of the proposed ensemble model has been compared separately against Support Vector Machine and Decision Tree and identified a statistical significance in performance.

4.1 Observation of outcome applying stacked ensemble model

A Stacked ensemble model has been developed by combining two diversified weighted ensemble classifiers. This model has been trained with the selected dataset from UCI, where feature selection has been carried out by applying LASSO. Accuracy, Sensitivity, Specificity, Precision, F1-score, and AUC Score achieved are 81.53%, 81.37%, 81.81%, 85.38%, 83.28%, 0.89, respectively. The generated confusion matrix and AUC-ROC curve have been shown in Figs. 9 and 10, respectively.

4.1.1 Applying K fold cross validation

10 Fold Cross Validation has been applied over the proposed ensemble model. The whole data set was randomly divided into ten approximately equal-sized folds, keeping the percentage of positive and negative class samples in each fold. In each iteration, nine folds were used for training the model. This procedure was repeated ten times so that each fold was used as the test set exactly once. The performance metrics are mean across all folds. The consecutive performances on each iteration has been shown in the Fig. 11 below.

4.1.2 Paired t test of the proposed model with other models

The paired t-test is used to test if the improvement in performance of your proposed model over other models is statistically significant over the same data splits. The performance of the proposed model has been tested against Support Vector Machine and Decision Tree. The experimental outcome has been shown in the Fig. 12 below.

4.2 Observation of outcome applying genetic algorithm over the stacked ensemble model

In this phase, the performance of the proposed ensemble classifier has been optimized by applying the genetic algorithm. Accuracy, Sensitivity, Specificity, Precision, F1-score, and AUC score have been optimized to 88.52%, 87.88%, 89.28%, 90.06%, 89.26% and 0.92% respectively. The confusion matrix and AUC-ROC curve have been shown in

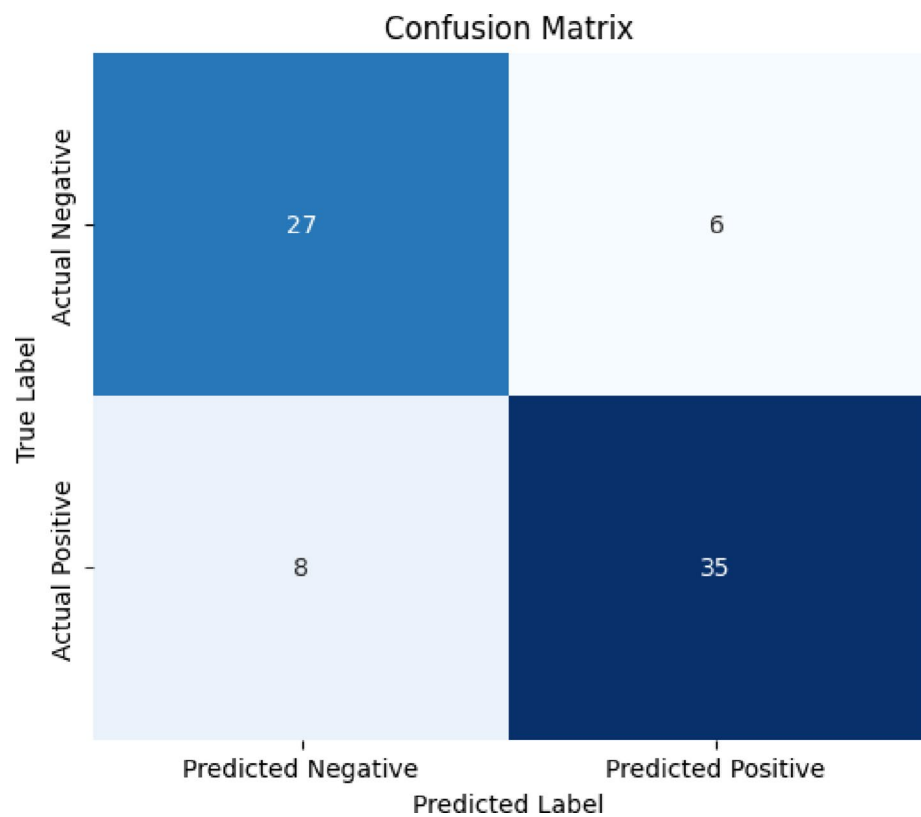


Fig. 9 Generated confusion matrix through the experimental result applying the proposed stacked ensemble classifier

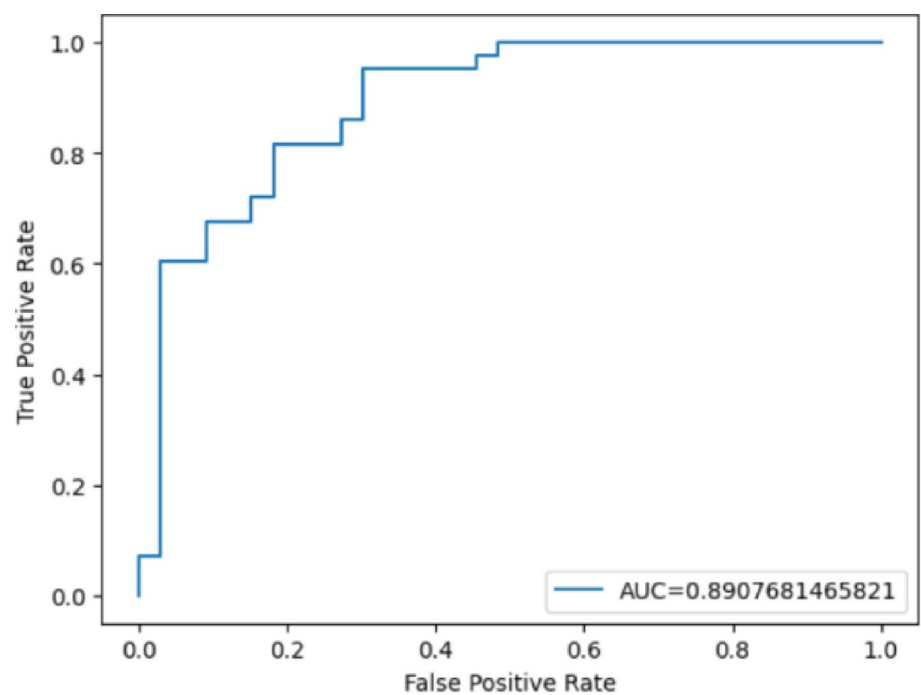


Fig. 10 Generated AUC-ROC through the experimental result applying the proposed stacked ensemble classifier

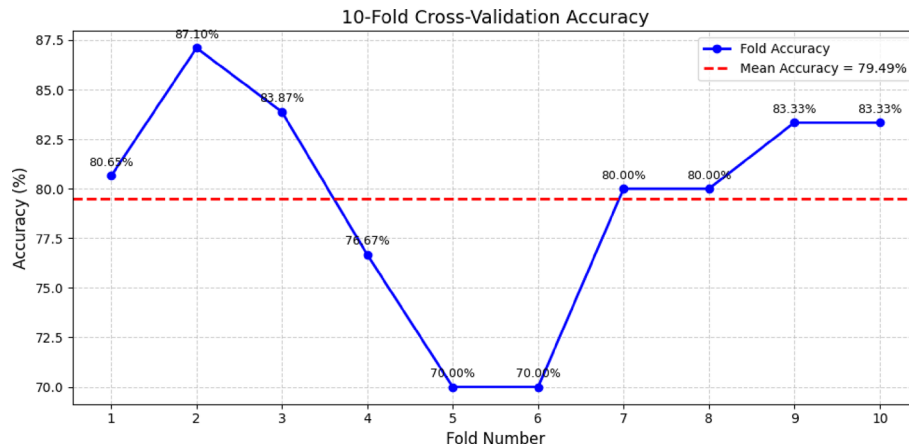


Fig. 11 Performance of the proposed model in 10 Fold cross validation

Comparison: Proposed Ensemble vs SVM

t-statistic = 6.2396

p-value = 0.0002

Result: Statistically Significant ($p < 0.05$)

Comparison: Proposed Ensemble vs Decision Tree

t-statistic = 2.8526

p-value = 0.0190

Result: Statistically Significant ($p < 0.05$)

Fig. 12 Paired t test outcome

Figs. 13 and 14, respectively. The achieved Cohen's Kappa Score of the proposed model is 0.769 and Matthews Correlation Coefficient (MCC) is 0.771.

The summarized performance of the proposed work in respect to metrics accuracy, sensitivity, specificity, precision and F1 score is shown in Fig. 15.

Accuracy shows how correct the model is overall by figuring out the percentage of cases that were correctly identified out of all samples. The proposed stacked model has achieved 81.53% accuracy while the optimized model has achieved 88.52% accuracy. Sensitivity, which is also called recall or the true positive rate, measures how well the model finds real positives by correctly identifying positive cases. Achieved sensitivity of the proposed stacked model and optimized stacked model are 81.37% and 87.88% respectively. The true negative rate, or specificity, measures how well the model avoids false alarms by correctly identifying negative cases. 81.81% and 89.28% sensitivity have been achieved by the proposed stacked model and optimized stacked model respectively. Precision measures how reliable positive predictions are by figuring out how many of the expected positive samples are actually positive. In this proposed worked 85.38% precision has been achieved by proposed stacked model and 90% precision has been achieved by optimization with Genetic Algorithm. The F1 score combines precision and sensitivity into one number by finding the harmonic mean of both. This gives a fair picture of how well the model is doing, especially when there is a class imbalance. 83.28% and

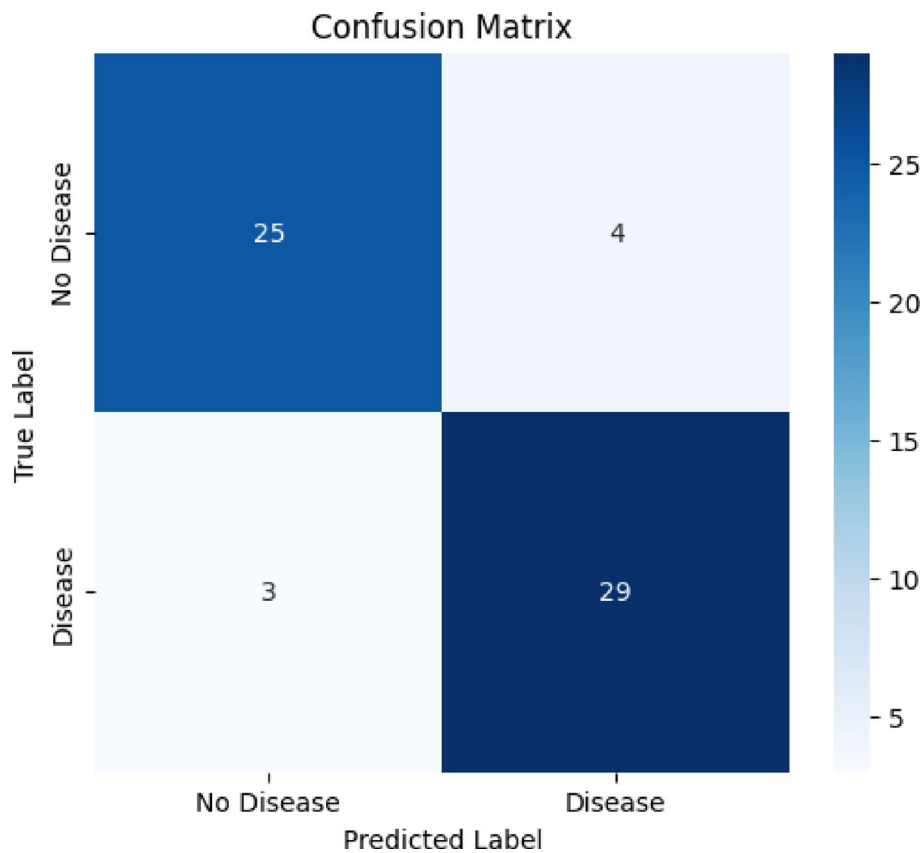


Fig. 13 Generated confusion matrix through the experimental result applying the genetic algorithm over the proposed stacked ensemble classifier

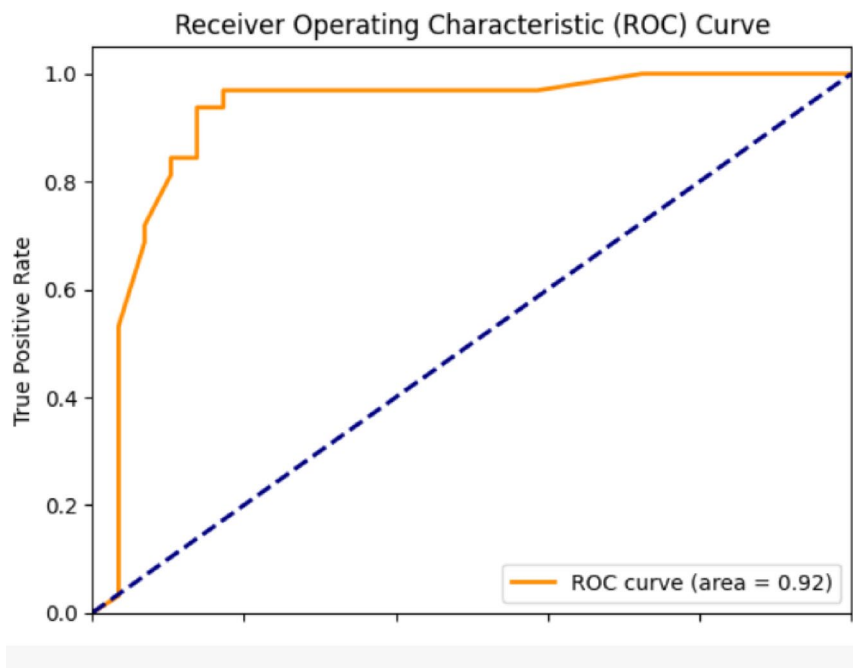


Fig. 14 Generated AUC-ROC through the experimental result applying the genetic algorithm over the proposed stacked ensemble classifier

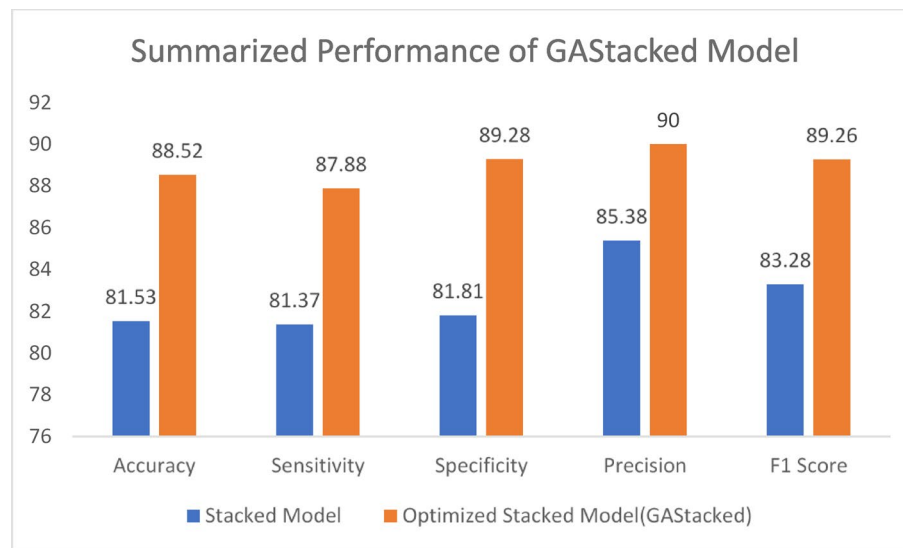


Fig. 15 Summarized performance of the proposed GA stacked model

89.26% F1 score has been achieved by the proposed stacked model and optimized stack model. The experimental outcome signifies that the classification are much better than random assignment, with good agreement between predicted and actual labels demonstrating dependable and resilient prediction.

These indicators, when used together, give a complete picture of a model's accuracy and reliability across many classes. This helps find both its strengths and weaknesses in making predictions. They are widely used in important fields like medical diagnosis, where both accurate detection and avoiding wrong predictions are very important.

Analysing the experimental results it has been identified that the application of the genetic algorithm over the proposed stacked ensemble classifier helped to achieve better performance. This proposed optimized algorithm can be utilized for early and accurate prediction of heart disease.

4.3 Experimental outcome after applying cross validation

In this phase of result analysis 10 Fold Cross Validation has been applied and performance metric –accuracy, precision, recall, F1 Score and AUC score distribution has been observed as shown in the Fig. 16 below.

4.4 Estimation of computational complexity of the proposed ensemble based GA model

Computational complexity of the proposed model has been calculated as Average Training Time (sec) = 2.139144 Average Prediction Time (sec) = 0.096771. Peak Memory used = 0.159681.

4.5 Model comparisons with some base line model

The proposed ensemble model has been compared with Support Vector Machine and Decision Tree and Artificial Neural Network classifier with 10 Fold Cross Validation with the parameter accuracy and computational complexities. It has been identified that the proposed model outperformed other models in respect to accuracy and outperformed ANN in respect to complexities.

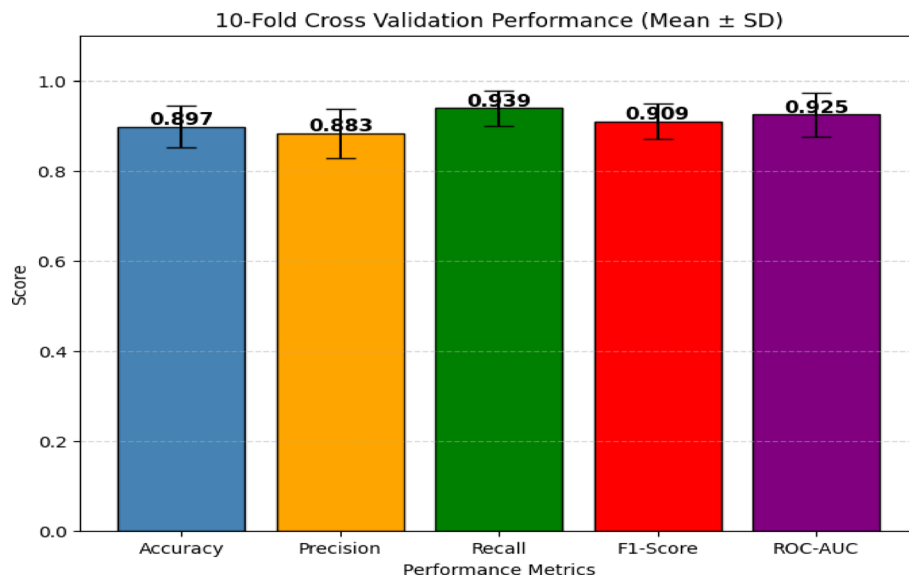


Fig. 16 10 Fold cross validation performance

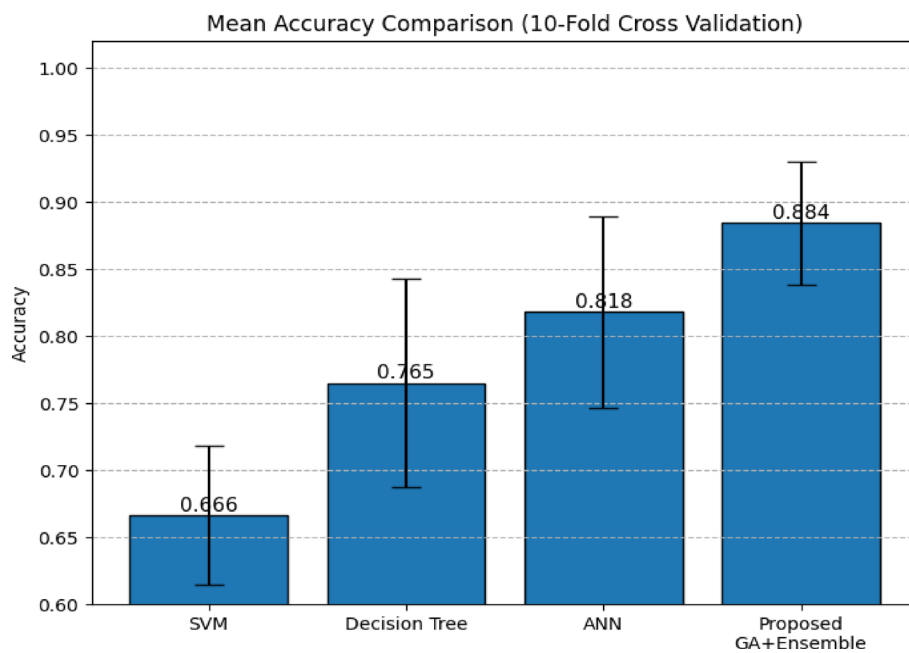


Fig. 17 Mean accuracy comparison

The experimental outcome with 10 Fold cross validation of Support Vector Machine, Decision Tree, Artificial Neural Network and Proposed Model has been shown in the Fig. 17 below.

The computational complexities comparison has been shown in the Fig. 18 below.

4.6 Model explanation using SHAP

The proposed model explanation has been tested using SHAP. Shapley Additive explanations leverages cooperative game theory to explain predictions of machine learning models. It looks at the marginal effect of each feature in all feature combinations to

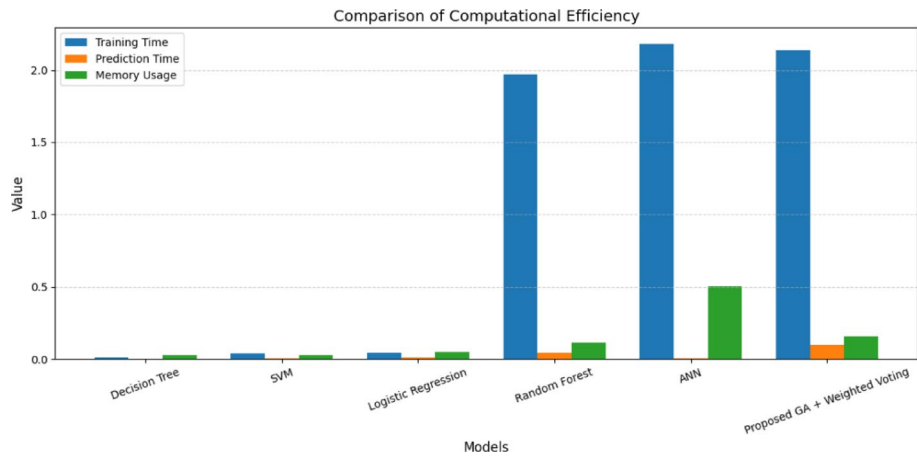


Fig. 18 Computational complexity comparison

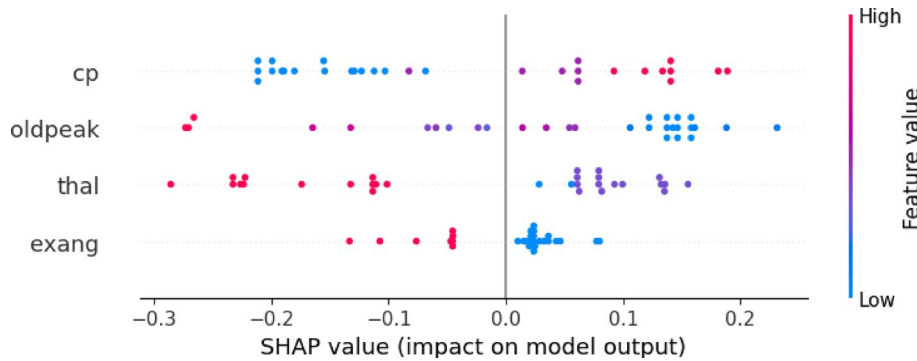


Fig. 19 SHAP value impact

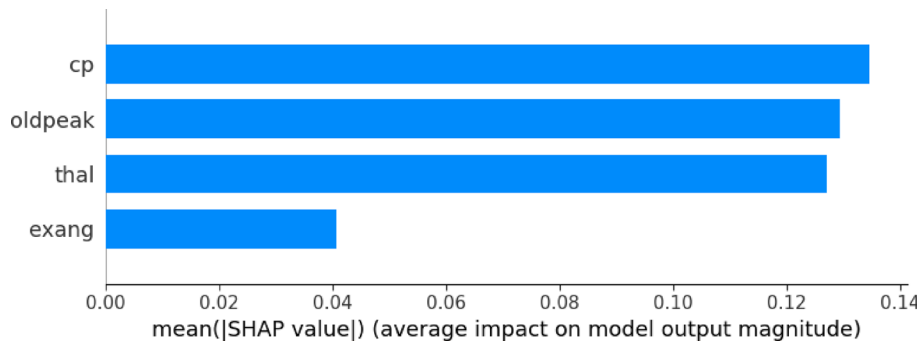


Fig. 20 SHAP average impact on model output magnitude

estimate its contribution to a single prediction such that the sum of all contributions is the final prediction minus the average of the dataset.

The experimental outcome that demonstrate the feature importance and contribution towards decision making has been shown in Figs. 19 and 20.

Overfitting is still a major problem in prediction models in healthcare. In the proposed framework, this problem is tackled by combining feature selection, Genetic Algorithm based optimization and ensemble learning. The goal of the Genetic Algorithm is to identify an optimal subset of features that maximises predictive performance and minimises

model complexity, thus reducing the chances of fitting dataset-specific noise. Ensemble learning also improves the generalisation by aggregating the predictions of several classifiers to reduce the variance and enhance the robustness. In addition, stratified k-fold cross-validation was used to assess the model on several data splits, maintaining the class distribution for training and testing. While these strategies significantly decrease the chance of overfitting.

A comparative analysis of the proposed work with few existing works in this domain has been presented in Table 1 below.

The performance of the proposed model has been tested against other three classifiers-Logistic Regression, Decision Tree and XGBoost optimized with Genetic algorithm. The experimental outcome has been presented in the Fig. 21 below.

This proposed heart disease prediction model has exhibited strong performance with an accuracy of 88.52%, suggesting its capacity to correctly anticipate outcomes in a high percentage of cases. Its efficiency in detecting people with cardiac disease is demonstrated by its 87.88% sensitivity, while its accuracy in classifying individuals without the ailment is demonstrated by its 89.28% specificity. An additional indication of the

Table 1 Comparative analysis of the proposed work with few existing works

Method	Observations
[1]	Applied KNN for disease classification and CNN for feature selection
[2]	Applied Naïve Bayes, KNN, DT, Random Forest, and SVM Average accuracy achieved: 75–90%
[3]	Random Forest, LSTM, and SVM have been applied SVM is employed to determine the potential ailment
[6]	RUSBoost has been applied to alleviate the problem of class imbalance
[9]	DNN with LightGBM and XGBoost machine learning models have been applied to predict diseases using laboratory test results Accuracy: 92% F1-score: 81%
[11]	RF, XGB, MLP, ARIMA, ARIMAX, GLARMA, and SARIMA have been applied
[13]	Recursive Neural Network and Long Short-Term Memory (LSTM) have been applied Accuracy Achieved: 89%
[21]	Temporal Association Rules (TARs) were employed in a Naïve Bayes classification model Accuracy achieved: 82%
[22]	Principal Component Analysis has been applied AUC:0.87
[24]	Ensemble Learning has been applied for classification, and the Chi-Square Test has been applied for feature selection Accuracy achieved:85.52%
[27]	Random Forest, Regression models, K-nearest Neighbor imputation, and Naïve Bayes algorithm have been applied Accuracy achieved:88.52%
[28]	Naïve Bayes, Decision Tree, Support Vector Machine, Logistic Regression, K Nearest Neighbour, and Random Forest have been applied Average AUC: 0.88
[29]	A hybrid Random Forest with Linear Model has been applied Accuracy achieved: 88.7%
Proposed GA Stacked model	Applied the stacked ensemble model and genetic algorithm Achieved accuracy:88.52% Sensitivity:87.88% Specificity:89.28% Precision:90% F1-score:89.26% AUC: 0.92

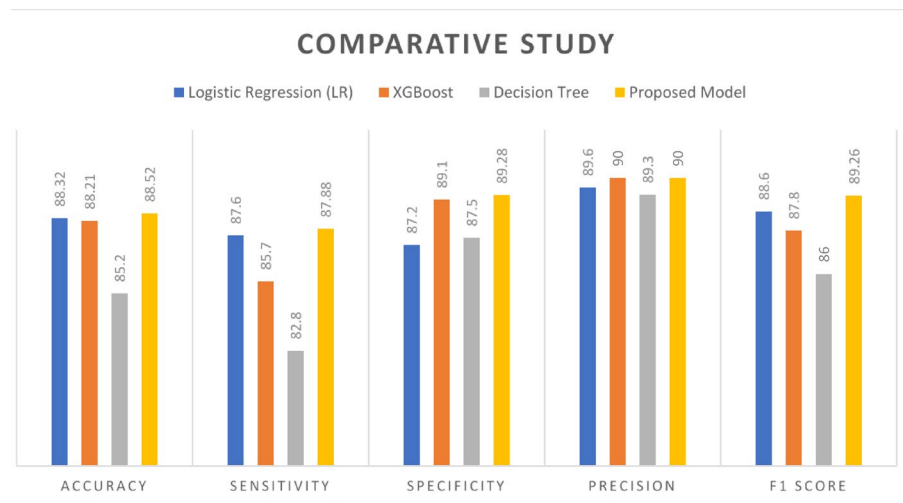


Fig. 21 Performance comparison of the proposed model with three baseline models across key evaluation metrics

model's balanced performance in managing false positives and false negatives is its 90% precision and 89.26% F1-score. The model's robust discriminatory power to distinguish between healthy and ill people is demonstrated by its AUC of 0.92. The achieved Kohen's Kappa score of the proposed model is 0.679 and Matthews Correlation Coefficient (MCC) = 0.771.

The value of this model's contribution to the prediction of heart illness lies in its ability to help medical practitioners identify and treat heart problems early on, potentially lessening the severity of heart-related issues.

5 Conclusion

With an accuracy of 88.52%, the two-layer customized ensemble model introduced in this study has demonstrated itself to be a successful method for predicting cardiovascular illness. The predictive power of the model has been improved by combining base learners (SVM, Decision Tree, Random Forest, and Logistic Regression) with LASSO feature selection and optimization based on genetic algorithms. The proposed framework has contributions in terms of novel integration of evolutionary feature optimisation, hierarchical weighted ensemble learning, statistically validated robustness and comprehensive performance evaluation. These properties improve the practical applicability and robustness of the proposed heart disease prediction decision-support framework.. The proposed framework relies solely on structured clinical attributes and does not leverage other clinically relevant data sources, such as medical images, electrocardiograms, genomic information, wearable sensor data, or longitudinal electronic health records. In future work, the framework will be validated on different clinical datasets, ensemble weights will be optimised by using adaptive optimisation techniques, computational complexity will be reduced, and the model will be extended to multimodal healthcare data to improve its clinical applicability and robustness.

Author contributions

A.B.M. and S.G. contributed to the conceptualization and design of the study. S.M. and D.S. were responsible for data curation, methodology, and implementation of the models. B.A. contributed to supervision, validation, and critical revision of the manuscript. A.D.D. was involved in analysis, interpretation of results, and manuscript drafting. All authors reviewed and approved the final manuscript.

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Data availability

The dataset used in this study is publicly available from the UCI Machine Learning Repository. Specifically, the Heart Disease Dataset can be accessed at: <https://archive.ics.uci.edu/dataset/45/heart+disease>. The dataset is also identifiable via the <https://doi.org/10.24432/C52P4X>.

Declarations**Ethics approval and consent to participate**

Not applicable.

Consent for publication

Not applicable.

Clinical trial number

Not applicable.

Competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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